Thapar Institute of Engineering and Technology
Mid Semester Examination (Odd 2425)
MCA302- Object Oriented Programming

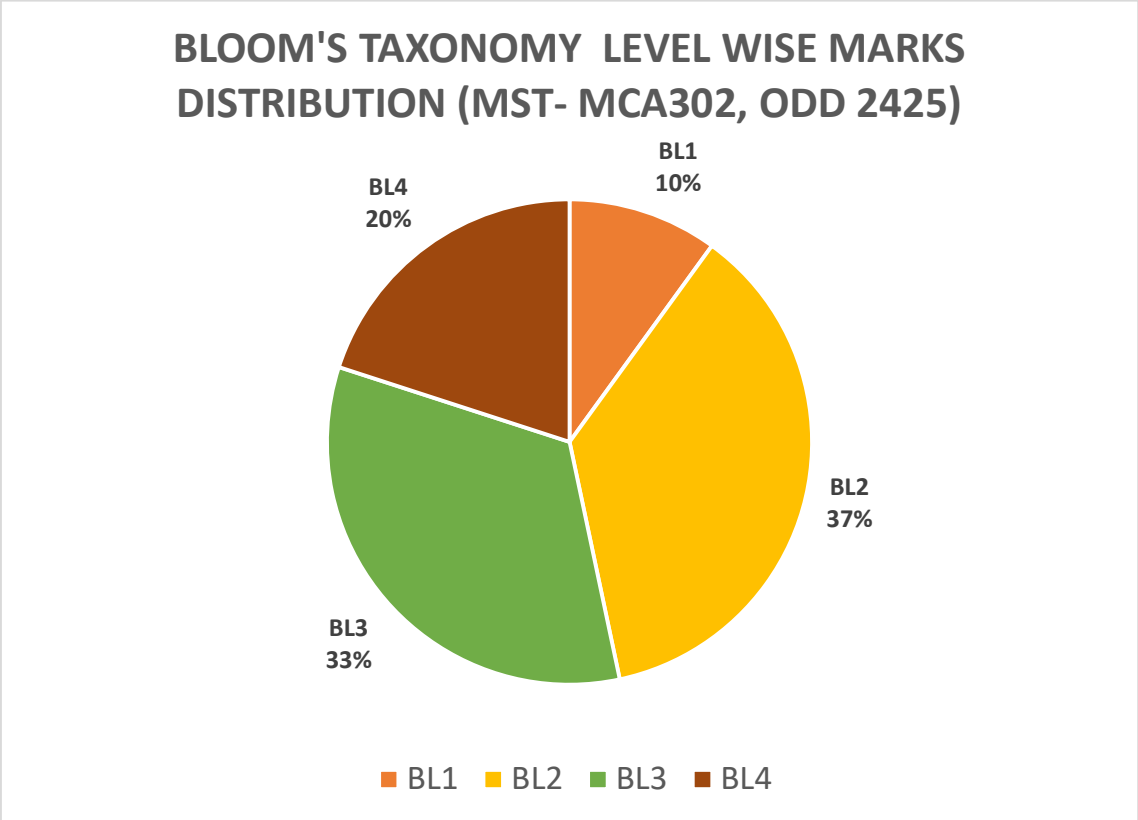
Date: 25/Sept/2024, 11:00 Am **Instructor: Dr. Jaskirat Singh**

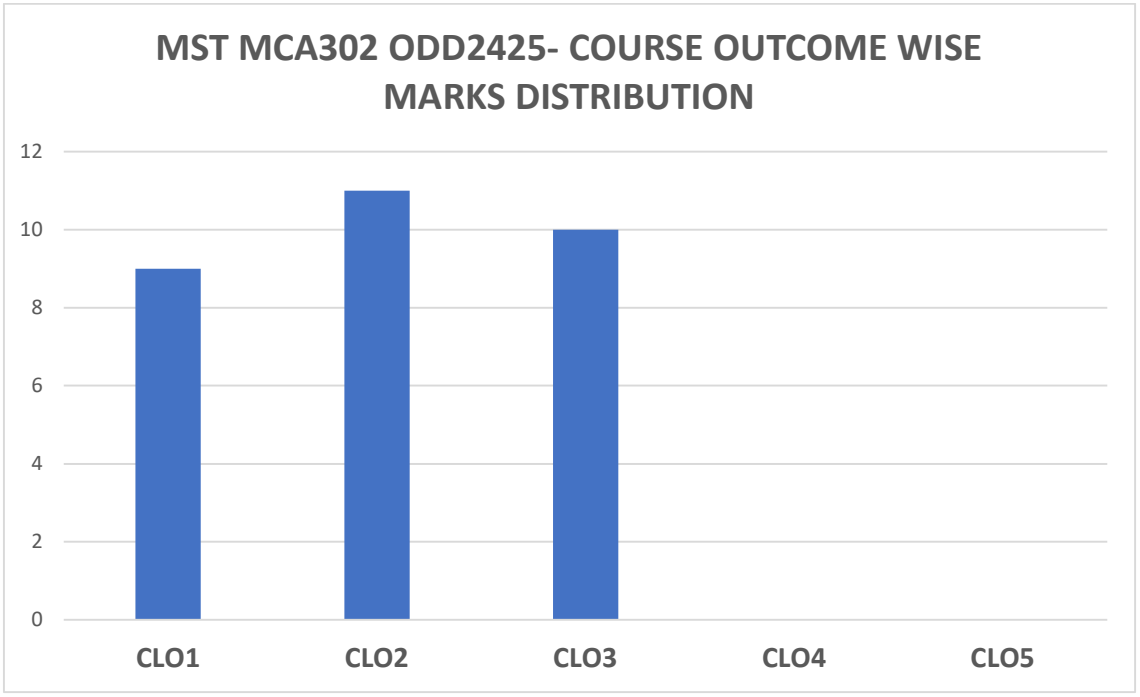
Total Time: 2 hours **Total Marks: 30**

Instructions: *Attempt all Questions. Attempt all questions in sequence and all parts of the question together. Give output for the programs written by you.*

Q.NO	QUESTION	Marks	CLO	BL
1a.	You are developing a software system for the Thapar Institute of Engineering and Technology (TIET) to manage and track the total number of departments in the institute. Each department is represented by a Department class, and every department has a unique name. TIET has a central administration system that needs to keep track of the total number of departments, and this information should be accessible to all department instances. Additionally, the institute has a special department, the "Administrative Department," which is responsible for overseeing all other departments. This department must be initialized automatically when the program starts and should remain active until the program ends. The Administrative Department should have a fixed name, and its lifecycle should be managed automatically by the program. **Requirements:** 1. Implement a data member in the Department class to keep track of the total number of departments at TIET. 2. Implement a member function that returns the total number of departments. 3. Create a object for the Administrative Department that is automatically created when the program starts and destroyed when it ends. 4. Ensure that the object for the Administrative Department does not interfere with the regular department count. **Question:**Write the C++ code with output for the best implementation of the Department class to fulfil the above requirements?	**6**	**1**	**L4**
1b.	Discuss the different scenarios in which naming conflicts can arise in C++ programming, leading to ambiguity in code. What are the common causes of these conflicts, and how can they be resolved? Include examples to illustrate your points.	**4**	**3**	**L2**

2a.	Write a C++ code with output that demonstrates the use of a pointer to access and modify a dynamically allocated Student object. Take student name and an array of marks to store marks of 3 subjects as data member. Implement a constructor and other suitable member function (for example: totalmarks()) that operate on the data members.	**4**	**2**	**L3**
2b.	Explain what is a dangling pointer?	**1**	**1**	**L1**
3a.	Explain the difference between passing object as value, passing object as reference and passing object as pointer with the help of small code snippets(function)/example.	**3**	**2**	**L2**
3b.	Explain how C++ is different from C Language?	**2**	**1**	**L1**
4	You have been hired as a developer to develop a C++ application for a banking system that includes a custom class named Account, which has attributes such as **accountNumber, holderName,** and **balance**. Explain how would you overload both the stream extraction operator (>>) and the stream insertion operator (<<) for the Account class.	**6**	**3**	**L3**
5	What is a move constructor? How it is different from copy constructor. Explain with the help an example.	**4**	**2**	**L2**




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